CleanQuest — Judges' Presentation

Meta Global AI Developer Hackathon · Indie track · **Build for Your World**

CleanQuest makes the invisible 2 a.m. cleaning shift visible — night cleaners prove their work with two photos, own a verified record that travels with them job to job, and get recognized instead of re-checked.

Every screenshot below was captured 2026-10-31 from the real app. Meta stack: **Muse Spark 1.3** (per-surface scorecards) + **SAM 3.1** (surface overlays) + **Muse voice transcription** + **Baseten Spark Open** as a logged second provider — plus an on-device MediaPipe + classical-vision rung that plays with no keys and no signal.

Indie track: creativity · technical ambition · whether it works

Vanilla JS · no build · no backend · offline-first

EN / ES / PL throughout

Demo-ready: one-tap Judge Demo + live loop + spectator board

How to score this — rubric map

Criterion (weight)	CleanQuest's answer	Where to see it
Impact 30% — who benefits, does it matter, could people use it	Non-technical night cleaners get portable proof of work; small operators get proof without re-checks; associations get training delivered at the moment of work. Consent-first, faces never identified, read-only appeals, top-10-only boards.	Run the app in Spanish for 20 seconds; open the inspection log; file a dispute
Meta AI Tooling 25% — beyond a basic integration	Spark scorecards grade each surface with confidence gating (low-confidence marked unverified, never smoothed over); SAM 3.1 overlays ground every claim in pixels; voice notes transcribed; Baseten Spark Open toggle with exportable per-run log; on-device EfficientDet-Lite0 + Pose Landmarker rung.	Results scorecard with provenance line; provider log export; one-tap Judge Demo
Demo 25% — working, holds up live	One-tap Judge Demo runs the real pipeline on sample photos in ~10 seconds; then run the live loop on a real room; spectator board projects the ranking.	Instructions screen → Judge Demo; `?spectator=1` on the big screen
Technical Depth 20% — sound, refined engineering	11k lines of dependency-free vanilla JS; hermetic self-test + persona suite; offline queue that re-scores on reconnect; 44px touch floors verified on 9 mobile screens + desktop; no fake data paths (demos excluded from ranking, samples labeled and ineligible).	Settings → run self-test; airplane-mode run; this report's appendix

Not another image analyzer

A one-shot classifier would fail the gate. CleanQuest is a closed loop, and every link is in the build:

- **Play** — check in with a QR, system-assigned high-impact spot, frame guide, in-app capture only.
- **Verify** — three-bar scorecard (Clean 50 / Technique 25 / Time 25), per-surface verdicts with confidence, SAM overlays, provenance line naming the model.
- **Appeal** — one-tap read-only dispute, 5-business-day promise. No person anywhere can edit a score — enforced in code, not policy.
- **Record** — geofenced, exportable inspection log; portable cleaner-owned passport, opt-in employer share.
- **Train & crew** — 30-card trilingual swipe deck with missed-topic repeat, weekly crew matchups, system-chosen MVP, 13-week consistency meter.

Say it on stage: *"The system scores, nobody edits — here is the appeal button that proves it."*

Live demo script (2 runs, ~3 minutes)

1. **Run 1 — Judge Demo (10 seconds).** Instructions screen → *Judge Demo*. Narrate: *"This is a synthetic dirty/clean pair so you can see the scoring story without us mopping the judging room — same on-device pipeline, watch the provenance line."* Tap through to the scorecard; point at the three bars and the demo label.
2. **Run 2 — live loop.** Check in, shoot a real before/after in the room, finish, show the scored card with its provenance line ("Verified by …" with cloud keys pre-entered, or the on-device line without).
3. **Throughout:** spectator board (`?spectator=1`) projected; host voice explicitly enabled for run 2 (it defaults to Off — say why: occupied buildings). Keep a 30-second backup recording of a full loop in case the room, camera permission, or venue wifi fails.
4. **Before stage:** pre-enter cloud keys in the demo browser, pre-warm MediaPipe on venue wifi (models cache after one load), print area QR stickers, clear localStorage so no prior runs pollute the board.

How the AI works

CleanQuest runs a **two-rung vision ladder** — real on-device perception with no keys and no backend, plus an opt-in cloud rung that verifies and explains. One honest caveat: the classical scoring layer works with zero signal, but the MediaPipe weights fetch from CDN on first use and then cache — a first-ever run on airplane mode scores via pixel-diff plus the neutral pose fallback, with full ML perception on every run after.

Rung 1 — on-device (private by construction, ML after first load)

- **Object counting:** MediaPipe EfficientDet-Lite0 (int8 TFLite, loaded on demand, never bundled) counts objects before/after; a matcher measures "objects cleared". Score threshold 0.35, up to 30 results.
- **Change + structure:** classical vision on a downscaled frame — pixel difference (threshold 38), edge-based structural difference, and a 3×3 zone texture grid that also drives the live zone map.
- **Victory-pose style score:** MediaPipe Pose Landmarker Lite (CPU delegate, single pose) reads 33 landmarks into a 0–10 style score. No model, no person, no camera? It says so plainly and applies a neutral 7.5.
- **Why it matters:** photos never leave the browser on this rung; faces are never identified; no facial recognition or biometrics anywhere.

Rung 2 — cloud verification (opt-in, keyed, explained)

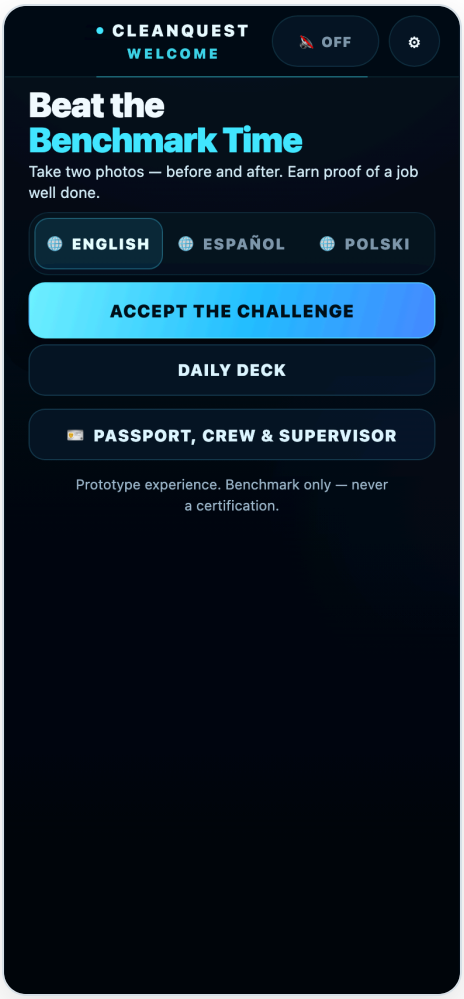
- **Verified scorecards:** `muse-spark-1.3` grades each surface (counter, glass, floor, fixture). Surfaces under 0.6 confidence are marked *unverified*.
- **Surface overlays:** SAM 3.1 segments each surface so the scorecard points at the pixels it means.
- **Voice notes:** transcribed via Muse voice transcription; without a key the audio attaches untranscribed.
- **Second opinion:** Baseten Spark Open toggle (`meta-models/muse-spark-open-1026`) with an exportable provider log.
- **Graceful failure:** no key or cloud down → the game plays on with an "Offline score" note; queued runs marked "awaiting score" are re-scored with the real pipeline on reconnect; a failed cloud check falls back to the on-device score with the failure stated on screen.

The host voice

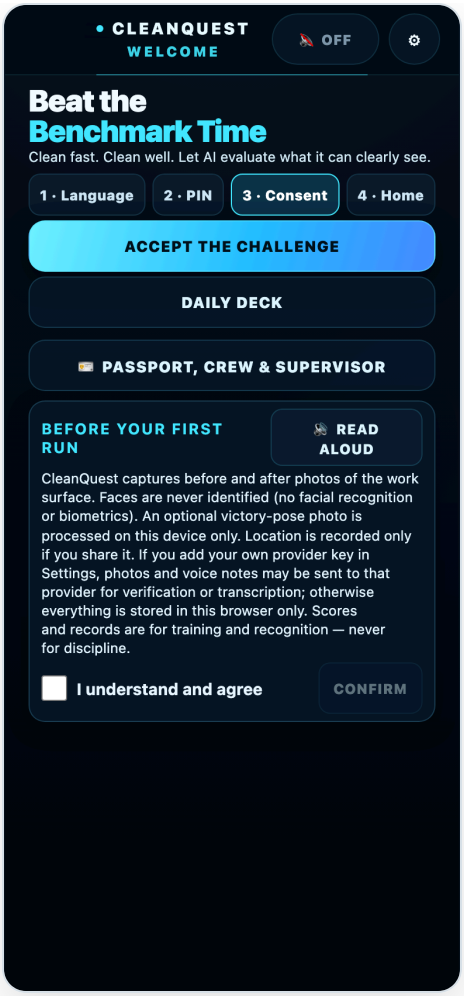
231 pre-rendered EN clips (ES/PL via TTS + synth) across verdict tiers routed by margin; scores spoken digit-by-digit so audio always matches the number; one stoppable WebAudio chain; **default Off, never plays uninvited** — including UI tones, which respect the mute.

Feature tour — what was built, how it looks

Welcome + onboarding



Entry hub: trilingual picker, Accept the Challenge, Daily Deck, Passport/Crew/Supervisor hub. First tap opens plain-language consent (photo use, no face ID, local-only) with read-aloud, plus optional phone + 4-digit PIN. READY

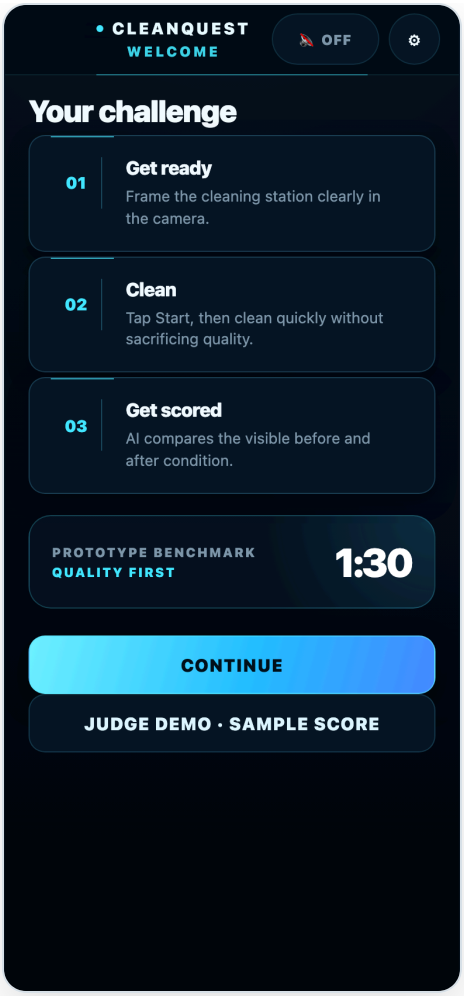


Consent-open state keeps every CTA ≥44px; CONFIRM is visible without scrolling. All gating chips verified at 44px.



Full ES translation holds layout — the workforce reads first, not translated afterward.

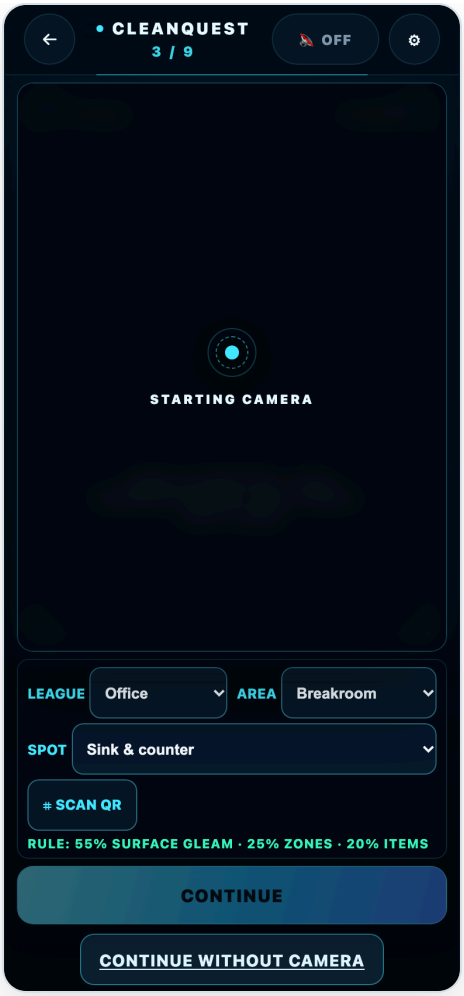
Instructions + one-tap Judge Demo



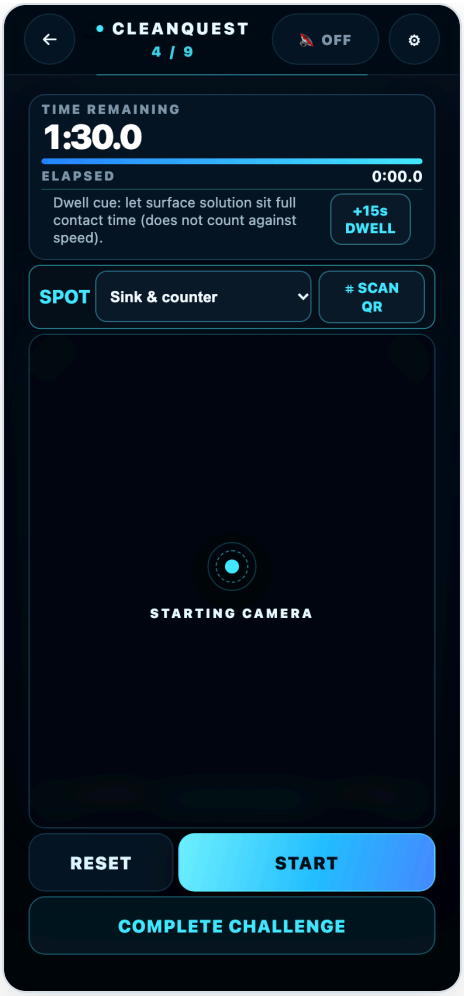
Camera check-in + timer

Three numbered steps, benchmark card, consent-gated camera entry — and the **Judge Demo** button: one tap runs the real local pipeline on sample photos and lands on a labeled scorecard. Consent-gated exactly like the real path.

READY — the 10-second scoring story

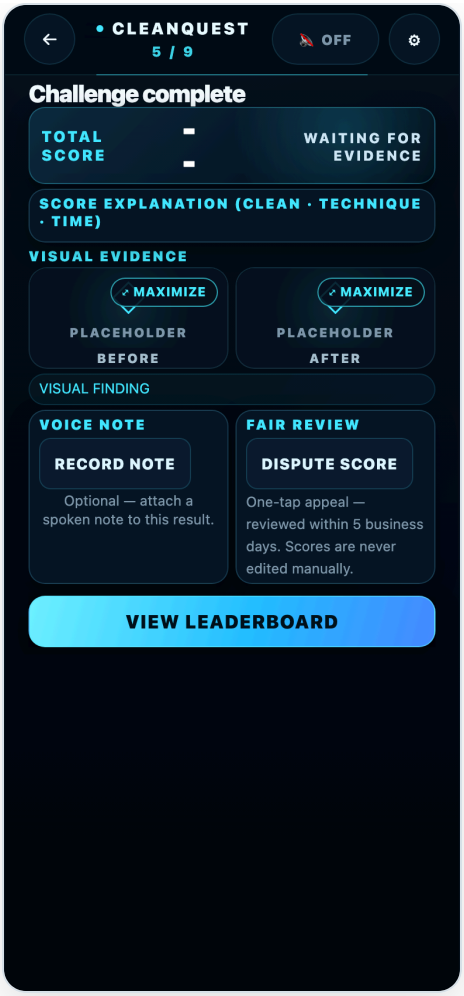


League / Area / system-assigned spot, QR scan, frame guide, in-app capture only. Short-viewport guard keeps CTAs in the first fold at 360×740. **READY (needs a real device to play)**



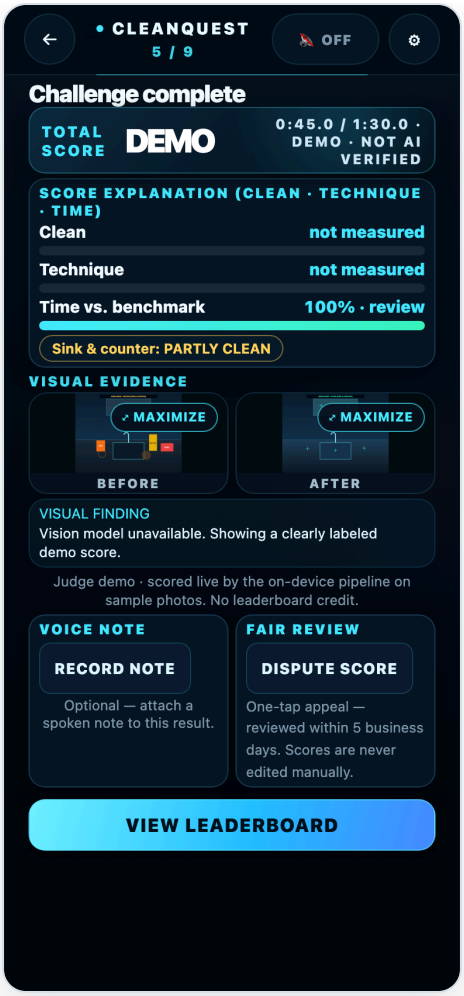
Countdown, ghost overlay to match the angle, dwell cue, host callouts, big Finish. Live zone grid stays neutral during the run — misses remain hidden until scoring, per the operating standards.

Results scorecard



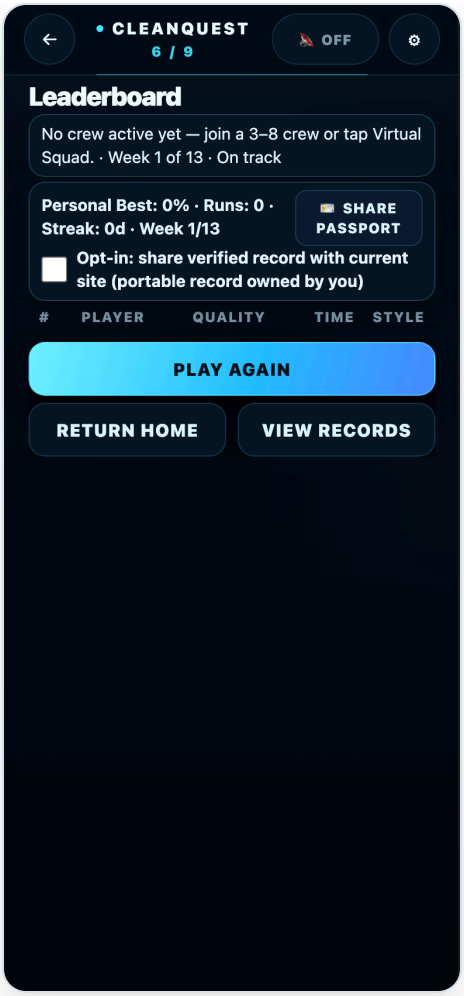
Weighted TOTAL hero (50/25/25), three explanation bars, honest surface map, uncropped evidence with lightbox, one-tap read-only Dispute, optional voice note, pose score — plus the **provenance line** (provider + model + SAM status, or the on-device line) and the **Judge Demo label** on demo runs.

READY



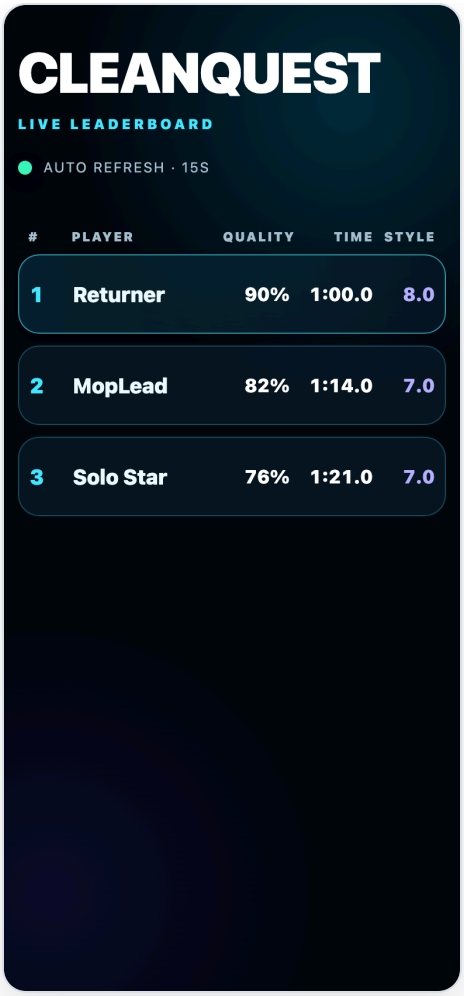
The Judge Demo result: real pipeline output on sample photos, labeled demo, no leaderboard credit, nothing persisted. (Headless capture shows the honest model-unavailable fallback; on a device this is a computed score.)

Leaderboard + spectator



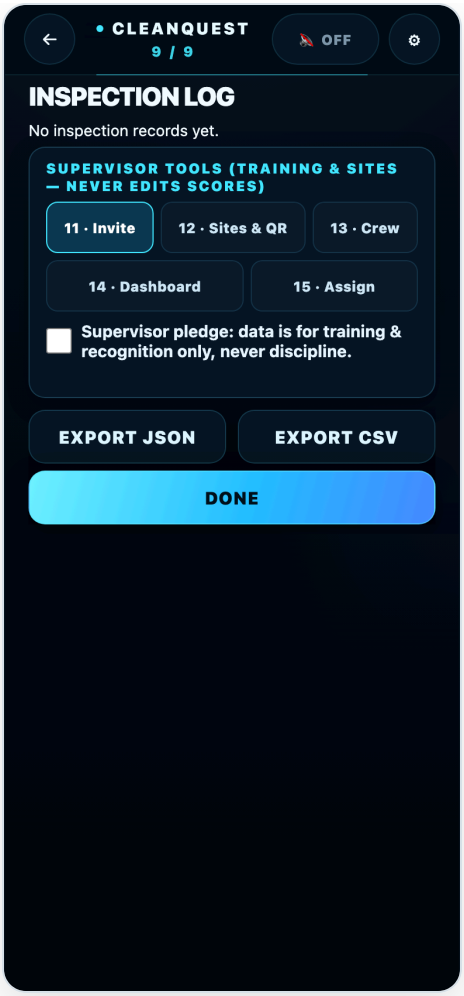
Top-10 + your own row only, practice and demo runs excluded, labeled sample rows excluded from ranking, personal best computed from eligible runs only. Fresh installs show an honest empty board — clear localStorage before judging.

READY — best demo surface

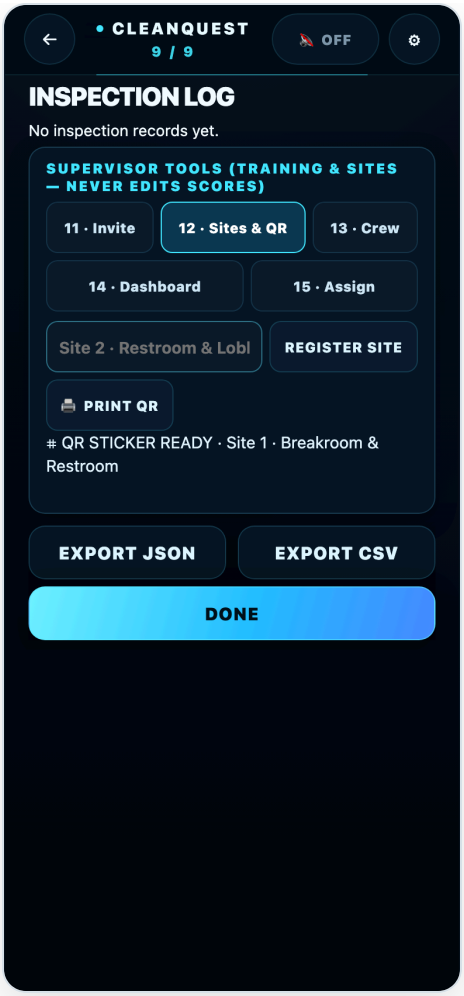


`?spectator=1` distance board, 15s refresh — project it on the big screen while someone plays.

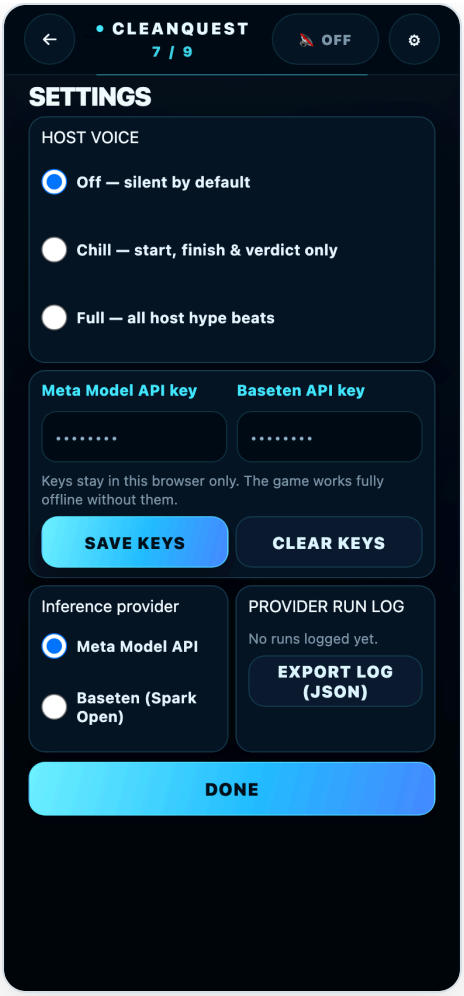
Records, settings, deck



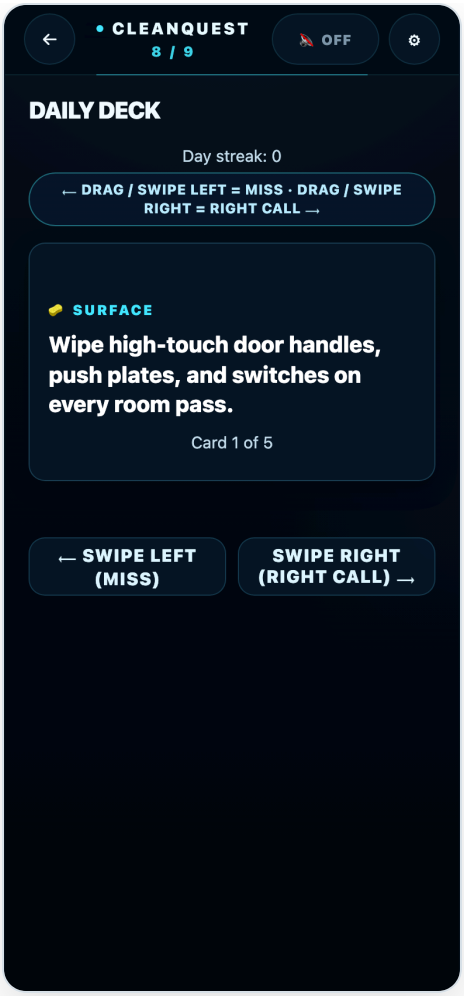
Geofenced inspection log with JSON/CSV export; supervisor panes 11–15 (pledge-gated tools, sites + printable QR stickers, bulk invite, training gaps, one-tap packs). Offline queue re-scores on reconnect with the real pipeline.



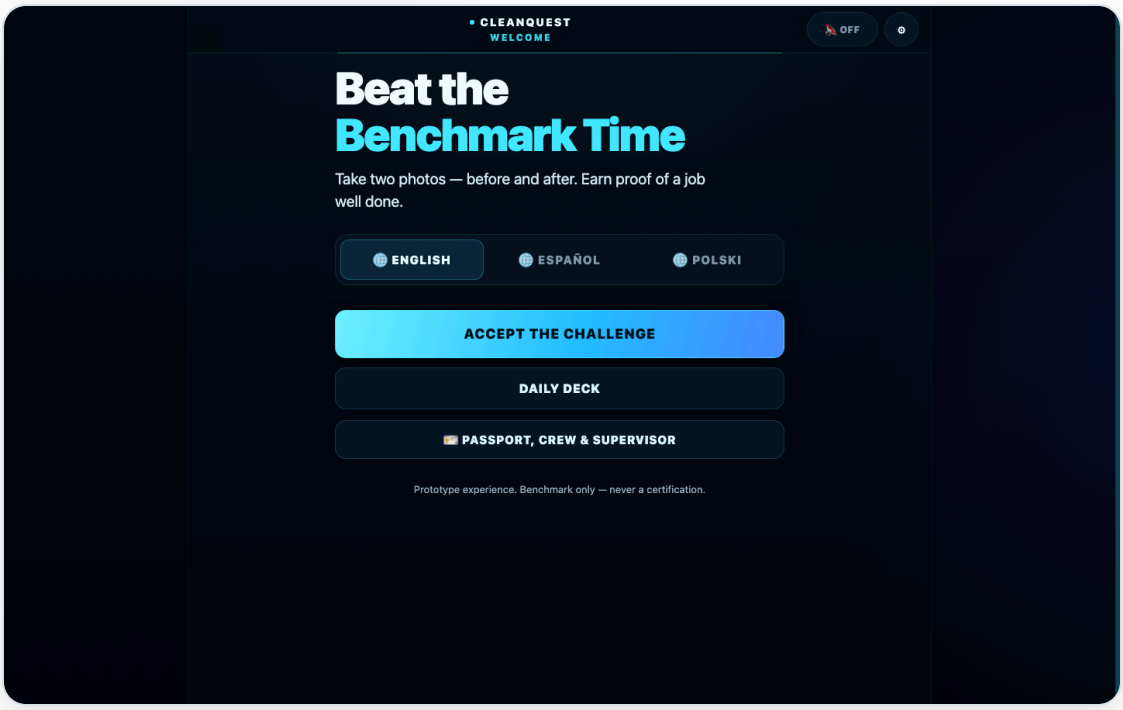
Sites pane with QR sticker preview — the physical-to-digital bridge.



Host voice control (default Off), API keys (browser-only), provider toggle + exportable run log, voice/test buttons.



30 trilingual technique cards, 5/day, missed-topic repeat, streaks, labeled sponsor slot after answers only.



Desktop holds up too: centered column, all CTAs in-fold at 1440×900, camera grid reflows to two columns. Mobile-first, desktop-clean.

Known limitations (disclosed, not hidden)

- **Registered-site prize gate:** eligibility today = system-assigned spot + in-app capture + practice exclusions. Matching runs against admin-registered sites needs a site-picker/QR-v3 change — scoped, not yet built.
- **Headless ≠ device:** MediaPipe needs real WebGL; CI/headless runs show the honest fallback path. Judge on a real phone.
- **Cloud needs keys:** without operator keys judges see the on-device path only — which is itself a complete game, with the offline note shown.
- **Residual detector advisory:** one low 11px-label note in dense game chrome; body copy is 12–14px+.

Appendix — verification

- Layout audit: 9/9 mobile screens at 360×740 (zero sub-44px layout targets, zero h-scroll, primary CTAs in-fold) + desktop 1440×900 pass (zero sub-44, zero h-scroll).
- Independent judge + advisor reviews run against the operating standards; all blockers fixed and re-verified (forced-unmute removed and tones mute-gated; live zone spoilers neutralized; judge-demo cloud egress blocked; demo/sample runs excluded from ranking, personal best, and sponsor counts; offline queue re-scores on reconnect).
- `node --check` clean; detector 8 → 1 low advisory; no secrets in repo; banned-token grep clean.
- Report screenshots are the app itself (390×844 mobile, 1440×900 desktop), captured 2026-10-31; leaderboard/spectator demo rows disclosed.